

Artificial Intelligence **IN MAKING FOOD!**

Food pairing- The theoretical number of recipes that can be generated is astronomically large. Even with a conservative estimate, one gets 10^{30} recipes with an average recipe size of 10 and 1,000 available ingredients.

Park et al. (2019) proposed a model called *KitcheNette* to determine the food-ingredient pairings scores and to predict the unknown pairings. Towards



this objective, they used MIT's Recipe1M database. *KitcheNette* utilized Siamese neural network and indepth network to obtain maximum performance in predicting unknown pairings. Pointwise mutual information score, the ratio of probability of two ingredients occurring in a recipe to probability of ingredients occurring separately, was used to find good and bad pairings. A positive score indicated good pairing and a negative score, a bad pairing.

The pairings obtained by the *KitcheNette* model were compared with those from FlavorDB as a reference. Such data-driven models can propel the identification of compatible ingredients in future.

Taste prediction-To determine the redundant components, preprocessing was implemented using PCA (principal component analysis) and Boruta algorithm. Random Forest, Ridge Logistic Regression and Adaboost algorithms were used to predict the taste of molecules.



Novel recipe generation-For ages chefs and culinary enthusiasts have attempted to create novel recipes by tweaking ingredients and processing. Recent advances in computational gastronomy have led to the question, 'Can computers generate novel recipes?'

In one study conducted with a similar spirit, Lee et al. (2020) introduced an online pre-trained categories (low sugar, high fiber, low fat, grilling, and frying) to generate category vectors and probability of a category-specific recipe was computed using a nonlinear squash function.

Recently, Wang et al. (2020) proposed Decomposed Generation Network model, which generates cooking instructions by predicting structure of food images.



Agarwal et al. (2020) presented a Hierarchically Disentangled Model for text generation with named entities while curating 158,473 recipes using named entity recognition and unsupervised methods. The authors used two

models, the type prediction model, and the entity transformation model, which captured typical patterns among the same type of entities and then identified the most relevant entity. The output of the type prediction model was fed into the entity transformation model. This architecture can be used for ingredient-to-recipe generations, image-to-recipe generation, action-graphs-to-recipe generation, and translating a recipe from one language to another.

Morales-Garzo´ n et al. (2021) suggested an unsupervised method for adapting recipes. They used three datasets, one which they scraped to train word embedding model, second from food.com for validation of adaptation, and third was a nutritional database from COFID, containing 2,913 unique ingredients. These authors adapted a recipe in three ways: similarity-based (substitute ingredients based on similar recipes), preference-based (substitute ingredients based on user preferences), and restriction based (substitute ingredients based on user’s food constraints). Finally, recipes were evaluated through online surveys. This model did not implement any advanced neural language models such as BERT, Elmo or GPT.



Conclusion-Computational gastronomy is a new niche that food blends with data and computation. As a data and computation-intensive field it is rooted in well curated, structured data and computational algorithms for their investigation.

The computational gastronomy data resources on various aspects of food have been growing for the past few years and can have a more prospective future.